

HUDSON VALLEY, NY, USA

WMO: 725036

Lat: 41.626N

Lon: 73.882W

Elev: 166

StdP: 14.61

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14757

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.4	(c) 8.0	(d) -8.5	(e) 3.5	(f) 7.2	(g) -3.8	(h) 4.5	(i) 11.7	(j) 21.4	(k) 28.4	(l) 19.3	(m) 27.7	(n) 3.4	(o) 10	(p) 0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	91.2	73.6	88.4	72.3	85.7	70.9	76.4	86.7	75.0	84.1	73.5	81.9	8.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	124.1	81.3	72.2	119.7	80.2	70.8	114.1	78.6	39.9	86.8	38.5	84.0	37.2	82.3	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.6	(b) 16.8	(c) 14.4	(e) -5.7	(f) 96.5	(g) 6.4	(h) 3.8	(i) -10.3	(j) 99.3	(k) -14.1	(l) 101.5	(m) -17.6	(n) 103.7	(o) -22.3	(p) 106.4		
(a) 18.6	(b) 16.8	(c) 14.4	(e) -6.3	(f) 79.7	(g) 6.2	(h) 2.1	(i) -10.8	(j) 81.2	(k) -14.4	(l) 82.4	(m) -17.9	(n) 83.6	(o) -22.4	(p) 85.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	27.0	29.4	37.7	49.3	59.6	68.5	73.7	71.9	64.3	52.7	41.8	32.6	
	DBStd	18.10	10.73	9.63	9.32	8.30	7.52	6.35	5.33	5.37	6.92	7.80	8.44	8.88	
	HDD50	2685	715	579	395	109	8	0	0	0	1	61	272	545	
	HDD65	5936	1178	998	845	475	205	36	3	8	97	389	696	1006	
	CDD50	2983	2	1	15	89	305	554	736	678	429	144	26	4	
	CDD65	758	0	0	0	5	37	140	274	221	75	6	0	0	
	CDH74	7273	0	0	9	112	492	1372	2638	1934	652	62	2	0	
	CDH80	2480	0	0	2	31	152	451	1015	646	175	8	0	0	
Wind	WSAvg	5.1	5.6	5.9	6.5	6.2	5.2	4.6	4.2	3.7	3.9	4.7	5.2	5.2	
Precipitation	PrecAvg	44.1	2.9	2.6	3.2	3.7	4.4	4.0	4.4	3.9	3.9	3.6	3.6	3.3	
	PrecMax	61.2	8.3	5.7	7.8	8.9	12.4	8.8	14.4	11.6	12.6	18.2	8.6	9.2	
	PrecMin	26.0	0.5	0.4	0.2	1.1	1.0	0.3	0.7	0.7	0.5	0.4	0.7	0.6	
	PrecStd	8.5	1.7	1.1	1.5	1.6	2.5	2.1	2.8	2.2	2.5	3.0	1.9	1.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.9	61.7	73.4	84.2	89.6	93.0	96.6	93.5	89.9	80.5	70.3	63.4	
		MCWB	56.9	53.7	59.0	63.4	69.7	75.3	75.8	75.3	72.2	66.7	60.3	56.3	
	2%	DB	53.2	54.1	64.5	76.3	84.6	88.9	92.1	89.5	85.0	74.9	65.2	55.7	
		MCWB	49.0	45.8	53.0	59.1	67.5	72.3	74.2	73.5	70.2	63.9	58.4	51.3	
	5%	DB	45.9	48.1	58.4	70.3	79.8	85.1	88.9	86.3	81.1	71.0	61.2	51.3	
		MCWB	41.5	41.7	48.1	56.5	64.7	70.5	73.1	71.9	69.0	62.7	54.3	46.9	
	10%	DB	41.6	43.4	53.0	65.1	74.8	81.9	85.7	83.6	77.5	66.9	56.6	45.9	
		MCWB	37.3	37.5	45.6	53.1	62.3	68.8	71.4	70.5	67.0	59.2	50.5	41.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.8	56.2	61.0	66.5	73.2	77.1	79.0	78.4	75.7	70.1	63.9	58.2
			MCDB	60.1	60.9	71.2	79.5	86.0	89.4	91.8	88.9	83.7	76.2	68.0	61.6
2%		WB	48.8	47.4	54.9	62.4	69.8	74.9	76.7	76.0	73.3	67.2	59.7	52.1	
		MCDB	51.2	52.0	62.3	72.5	79.9	84.9	87.9	85.0	80.8	72.2	64.0	55.3	
5%		WB	42.2	42.0	50.2	58.6	67.3	73.0	75.2	74.5	71.2	63.9	55.7	46.8	
		MCDB	45.5	47.5	56.9	67.7	76.2	81.8	84.6	82.7	77.8	69.1	59.5	50.2	
10%		WB	37.8	38.2	45.9	54.9	64.5	70.9	73.6	72.9	69.3	60.3	51.4	41.9	
		MCDB	40.9	42.9	52.1	62.7	72.2	78.8	82.3	80.3	75.2	65.7	55.7	45.8	
Mean Daily Temperature Range	5% DB	MDBR	16.9	18.5	19.5	22.2	22.4	21.5	21.5	21.0	21.9	21.1	19.0	15.9	
		MCDBR	20.8	25.3	28.2	31.5	29.5	25.7	24.7	23.9	25.2	26.4	24.6	21.1	
		MCWBR	17.2	18.8	17.8	16.8	14.5	12.0	9.8	10.2	12.8	16.2	17.9	17.9	
	5% WB	MCDBR	20.0	22.7	25.4	26.8	25.0	21.8	21.0	19.9	20.5	22.0	21.1	19.9	
		MCWBR	17.5	18.0	18.1	16.5	13.4	11.4	9.8	9.6	11.7	15.7	17.6	18.0	
Clear-Sky Solar Irradiance	taub	0.305	0.323	0.348	0.370	0.413	0.471	0.483	0.461	0.386	0.356	0.323	0.304		
	taud	2.403	2.367	2.383	2.374	2.280	2.153	2.132	2.184	2.370	2.462	2.508	2.490		
	Ebn at Noon	267	279	285	286	274	258	253	255	269	265	260	259		
	Edh at Noon	25	31	34	37	42	48	48	44	34	28	23	22		
All-Sky Solar Radiation	RadAvg	561	839	1157	1471	1648	1791	1858	1632	1345	896	622	458		
	RadStd	42	80	90	157	163	165	106	102	132	95	57	48		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	-0.54	N/A	N/A	+205	+113
Regional (61 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+194	+104

Nomenclature: See separate page